

SN74LVC1G38 Single 2-Input NAND Gate With Open-Drain Output

1 Features

- Latch-up performance exceeds 100mA per JESD 78, Class II
- ESD protection exceeds JESD 22
 - 2000V Human-body model (A114A)
 - 200V Machine model (A115A)
 - 1000V Charged-device model (C101)
- Available in the Texas Instruments NanoStar™ and NanoFree™ packages
- Supports 5V V_{CC} operation
- Inputs accept voltages to 5.5V
- Supports down translation to V_{CC}
- Maximum t_{pd} of 4.5ns at 3.3V
- Low power consumption, 10 μ A maximum I_{CC}
- ± 24 mA output drive at 3.3V
- I_{off} supports partial-power-down mode and back-drive protection

2 Applications

- AV receivers
- Blu-ray players and home theaters
- DVD recorders and players
- Desktop or notebook PCs
- Digital radio or internet radio players
- Digital video cameras (DVC)
- Embedded PCs
- GPS: personal navigation devices
- Mobile internet devices
- Network projector front-ends
- Portable media players
- Pro audio mixers
- Smoke detectors
- Solid state drive (SSD): enterprise
- High-definition (HDTV)
- Tablets: enterprise
- Audio docks: portable
- DLP front projection systems
- DVR and DVS
- Digital picture frame (DPF)
- Digital still cameras

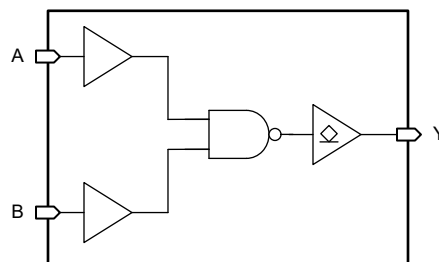
3 Description

The SN74LVC1G38 device is a single 2-input NAND gate. This device performs the Boolean function $Y = \overline{A} \cdot \overline{B}$ or $Y = \overline{A + B}$ in positive logic.

Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾	BODY SIZE ⁽³⁾
SN74LVC1G38	DPW (X2SON, 5)	0.8mm × 0.8mm	0.8mm × 0.8mm
	DRY (SON, 6)	1.45mm × 1mm	1.45mm × 1mm
	DSF (SON, 6)	1mm × 1mm	1mm × 1mm
	DCK (SOT-SC70, 5)	2mm × 2.1mm	2mm × 1.25mm
	DBV (SOT-23, 5)	2.9mm × 2.8mm	2.9mm × 1.6mm

- (1) For more information, see [Mechanical, Packaging, and Orderable Information](#).
- (2) The package size (length × width) is a nominal value and includes pins, where applicable.
- (3) The body size (length × width) is a nominal value and does not include pins.



Logic Diagram



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4 Pin Configuration and Functions

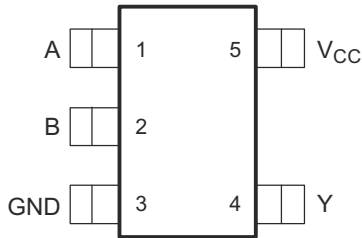
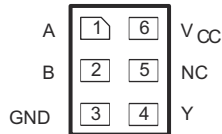
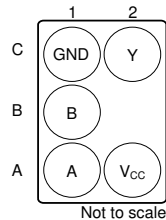


Figure 4-1. DBV Package 5-Pin SOT-23 Top View



NC – No internal connection.

Figure 4-3. DRY Package 6-Pin SON Top View



See mechanical drawings for dimensions

Figure 4-5. YZP Package 5-Pin DSBGA Bottom View

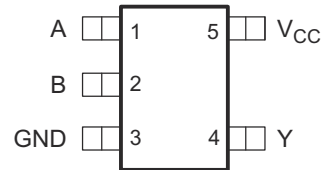


Figure 4-2. DCK Package 5-Pin SC70 Top View

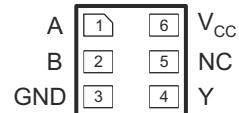


Figure 4-4. DSF Package 6-Pin SON Top View

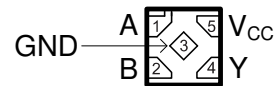


Figure 4-6. DPW Package 5-Pin X2SON Top View

Pin Functions

PIN				TYPE ⁽¹⁾	DESCRIPTION
NAME	DBV, DCK, DPW	DRY, DSF	YZP		
A	1	1, 5	A1	I	Logic Input A
B	2	2	B1	I	Logic Input B
GND	3	3	C1	—	Ground
NC	—	5	—	—	No Internal Connection
V _{CC}	5	6	A2	—	Positive Supply
Y	4	4	C2	O	Output Y

(1) I = input, O = output

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

			MIN	MAX	UNIT
V _{CC}	Supply voltage		−0.5	6.5	V
V _I	Input voltage ⁽²⁾		−0.5	6.5	V
V _O	Voltage range applied to any output in the high-impedance or power-off state ⁽²⁾		−0.5	6.5	V
I _{IK}	Input clamp current	V _I < 0		−50	mA
I _{OK}	Output clamp current	V _O < 0		−50	mA
I _O	Continuous output current			±50	mA
	Continuous current through V _{CC} or GND			±100	mA
T _J	Maximum junction temperature			150	°C
T _{stg}	Storage temperature		−65	150	°C

- (1) Operation outside the *Absolute Maximum Ratings* may cause permanent device damage. *Absolute Maximum Ratings* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Conditions*. If used outside the *Recommended Operating Conditions* but within the *Absolute Maximum Ratings*, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
- (2) The input negative-voltage and output voltage ratings can be exceeded if the input and output current ratings are observed.

5.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	2000
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	1000
		Machine Model (MM), A115A	200

- (1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT	
V _{CC}	Supply voltage	Operating	1.65	5.5	V
		Data retention only	1.5		
V _{IH}	High-level input voltage	V _{CC} = 1.65V to 1.95V	0.65 × V _{CC}		V
		V _{CC} = 2.3V to 2.7V	1.7		
		V _{CC} = 3V to 3.6V	2		
		V _{CC} = 4.5V to 5.5V	0.7 × V _{CC}		
V _{IL}	Low-level input voltage	V _{CC} = 1.65V to 1.95V	0.35 × V _{CC}		V
		V _{CC} = 2.3V to 2.7V	0.7		
		V _{CC} = 3V to 3.6V	0.8		
		V _{CC} = 4.5V to 5.5V	0.3 × V _{CC}		
V _I	Input voltage	0	5.5		V
V _O	Output voltage	0	5.5		V
I _{OL}	Low-level output current	V _{CC} = 1.65V	4		mA
		V _{CC} = 2.3V	8		
		V _{CC} = 3V	16		
			24		
		V _{CC} = 4.5V	32		
Δt/Δv	Input transition rise and fall rate	V _{CC} = 1.8V ± 0.15V, 2.5V ± 0.2V	20		ns/V
		V _{CC} = 3.3V ± 0.3V	10		
		V _{CC} = 5V ± 0.5V	5		
T _A	Operating free-air temperature	−40	125		°C

(1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. See [Implications of Slow or Floating CMOS Inputs](#).

5.4 Thermal Information

PACKAGE	PINS	THERMAL METRIC ⁽¹⁾						UNIT
		R _{θJA}	R _{θJC(top)}	R _{θJB}	Ψ _{JT}	Ψ _{JB}	R _{θJC(bot)}	
DBV (SOT-23)	5	357.1	263.7	264.4	195.6	262.2	-	°C/W
DCK (SOT-SC70)	5	371	297.5	258.6	195.6	256.2	-	°C/W
DRY (SON)	6	366.9	253.8	227.5	75.8	227.7	-	°C/W
DSF (SON)	6	406.2	201.0	256.9	35.2	256.6	-	°C/W
DPW (X2SON)	5	511.0	241.9	374.2	45.0	373.3	168.0	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application note.

5.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		V _{CC}	MIN	TYP ⁽¹⁾	MAX	UNIT
V _{OL}	I _{OL} = 100 µA	T _A = –40°C to +85°C	1.65V to 5.5V			0.1	V
		T _A = –40°C to +125°C					
	I _{OL} = 4mA	T _A = –40°C to +85°C	1.65V			0.45	
		T _A = –40°C to +125°C					
	I _{OL} = 8mA	T _A = –40°C to +85°C	2.3V			0.3	
		T _A = –40°C to +125°C					
	I _{OL} = 16mA	T _A = –40°C to +85°C	3V			0.4	
		T _A = –40°C to +125°C					
I _I A or B inputs	V _I = 5.5V or GND	T _A = –40°C to +85°C	1.65V to 5.5V			±1	µA
		T _A = –40°C to +125°C					
I _{off}	V _I or V _O = 5.5V	T _A = –40°C to +85°C	0			±10	µA
		T _A = –40°C to +125°C					
I _{CC}	V _I = 5.5V or GND, I _O = 0	T _A = –40°C to +85°C	1.65V to 5.5V			10	µA
		T _A = –40°C to +125°C					
ΔI _{CC}	One input at V _{CC} – 0.6V, Other inputs at V _{CC} or GND	T _A = –40°C to +85°C	3V to 5.5V			500	µA
		T _A = –40°C to +125°C					
C _i	V _I = V _{CC} or GND		3.3V		3.5		pF
C _o	V _O = V _{CC} or GND		3.3V		4.5		pF

(1) All typical values are at V_{CC} = 3.3V, T_A = 25°C.

5.6 Switching Characteristics, C_L = 15pF

over recommended operating free-air temperature range, C_L = 15pF (unless otherwise noted) (see Figure 6-1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	MAX	UNIT
t _{pd}	A or B	Y	T _A = –40°C to +85°C	V _{CC} = 1.8V ± 0.15V	2.9 7.4	ns
				V _{CC} = 2.5V ± 0.2V	1.7 3.8	
				V _{CC} = 3.3V ± 0.3V	1.5 4.9	
				V _{CC} = 5V ± 0.5V	0.9 2.4	

5.7 Switching Characteristics, C_L = 30pF or 50pF, –40°C to +85°C

over recommended operating free-air temperature range, C_L = 30pF or 50pF (unless otherwise noted) (see Figure 6-2)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	MAX	UNIT
t _{pd}	A or B	Y	T _A = –40°C to +85°C	V _{CC} = 1.8V ± 0.15V	2.8 10	ns
				V _{CC} = 2.5V ± 0.2V	1.6 6	
				V _{CC} = 3.3V ± 0.3V	1.4 4.5	
				V _{CC} = 5V ± 0.5V	1 3.9	

5.8 Switching Characteristics, $C_L = 30\text{pF}$ or 50pF , -40°C to $+125^\circ\text{C}$

over recommended operating free-air temperature range, $C_L = 30\text{pF}$ or 50pF (unless otherwise noted) (see Figure 6-2)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS		MIN	MAX	UNIT
t_{pd}	A or B	Y	$T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$	$V_{CC} = 1.8\text{V} \pm 0.15\text{V}$	2.8	11	ns
				$V_{CC} = 2.5\text{V} \pm 0.2\text{V}$	1.6	6.5	
				$V_{CC} = 3.3\text{V} \pm 0.3\text{V}$	1.4	5	
				$V_{CC} = 5\text{V} \pm 0.5\text{V}$	1	4.4	

5.9 Operating Characteristics

$T_A = 25^\circ\text{C}$

PARAMETER	TEST CONDITIONS		TYP	UNIT
C_{pd} Power dissipation capacitance	$f = 10\text{MHz}$	$V_{CC} = 1.8\text{V}$	3	pF
		$V_{CC} = 2.5\text{V}$	3	
		$V_{CC} = 3.3\text{V}$	4	
		$V_{CC} = 5\text{V}$	6	

5.10 Typical Characteristics

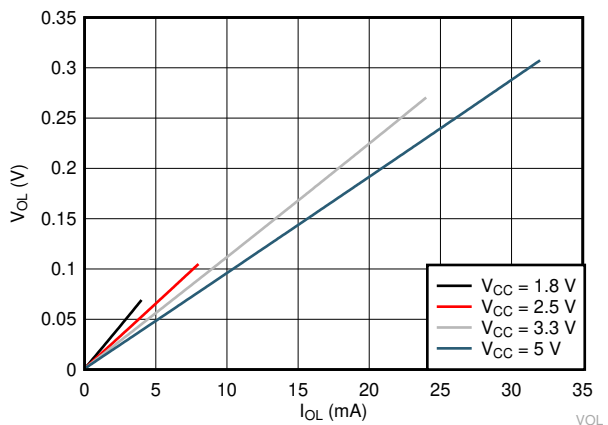
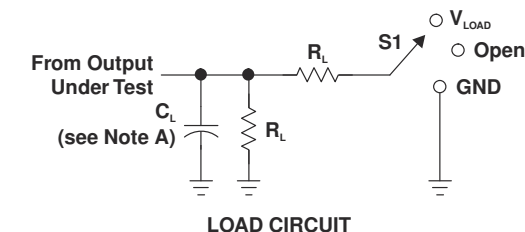


Figure 5-1. Typical I_{OL} vs. V_{OL} ($T_A = 25^\circ\text{C}$)

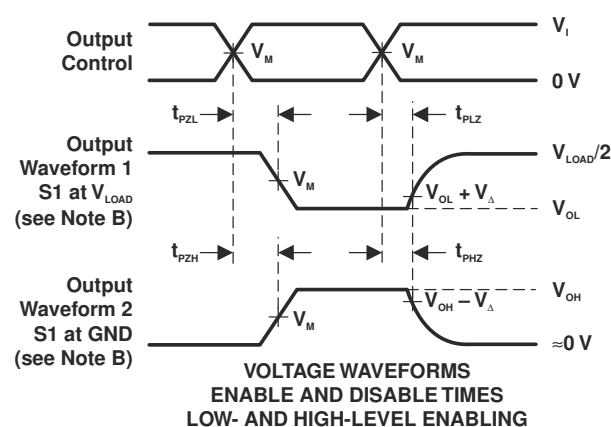
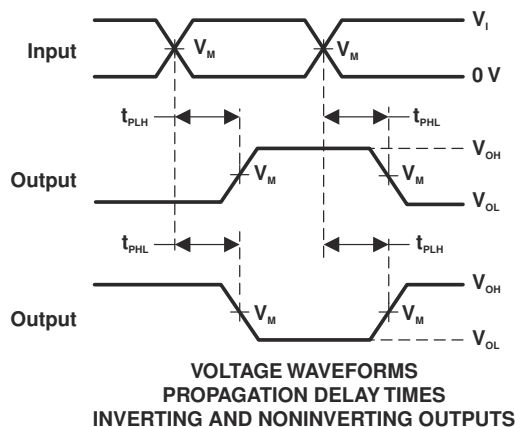
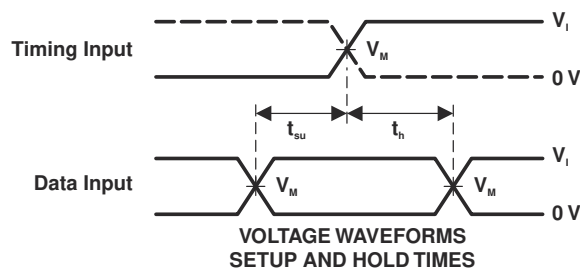
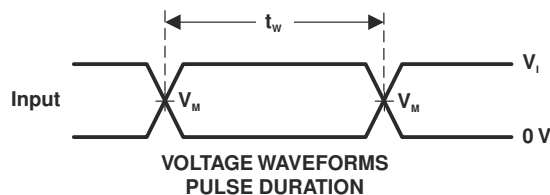
6 Parameter Measurement Information

(Open Drain)



TEST	S1
t_{PZL} (see Notes E and F)	V_{LOAD}
t_{PLZ} (see Notes E and G)	V_{LOAD}
t_{PHZ}/t_{PZH}	V_{LOAD}

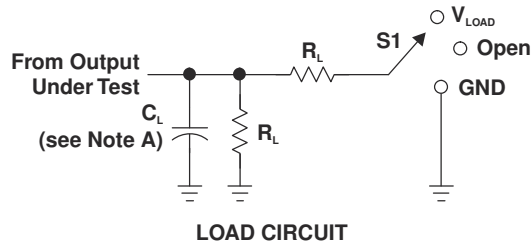
V_{CC}	INPUTS		V_M	V_{LOAD}	C_L	R_L	V_{Δ}
	V_i	t_i/t_f					
$1.8\text{ V} \pm 0.15\text{ V}$	V_{CC}	$\leq 2\text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	15 pF	1 M Ω	0.15 V
$2.5\text{ V} \pm 0.2\text{ V}$	V_{CC}	$\leq 2\text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	15 pF	1 M Ω	0.15 V
$3.3\text{ V} \pm 0.3\text{ V}$	3 V	$\leq 2.5\text{ ns}$	1.5 V	6 V	15 pF	1 M Ω	0.3 V
$5\text{ V} \pm 0.5\text{ V}$	V_{CC}	$\leq 2.5\text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	15 pF	1 M Ω	0.3 V



- NOTES:
- C_L includes probe and jig capacitance.
 - Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
 - All input pulses are supplied by generators have the following characteristics: $PRR \leq 10\text{ MHz}$, $Z_o = 50\ \Omega$.
 - The outputs are measured one at a time, with one transition per measurement.
 - Because this device has open-drain outputs, t_{PLZ} and t_{PZL} are the same as t_{PD} .
 - t_{PZL} is measured at V_M .
 - t_{PLZ} is measured at $V_{OL} + V_{\Delta}$.
 - All parameters and waveforms are not applicable to all devices.

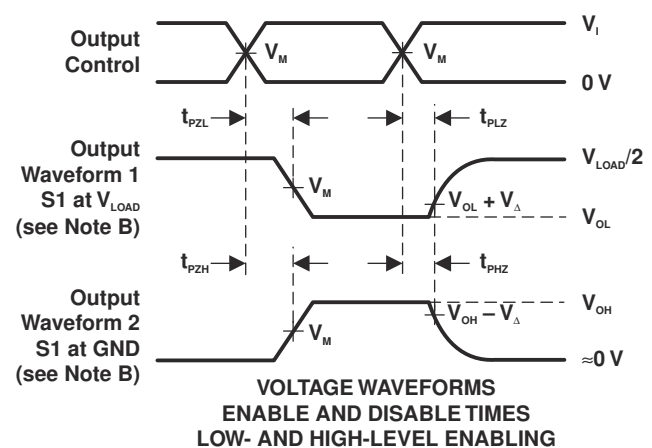
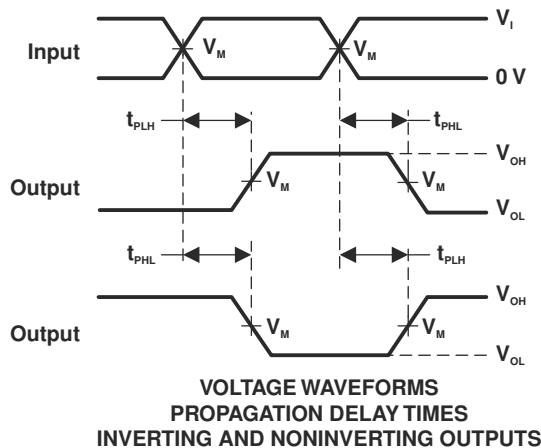
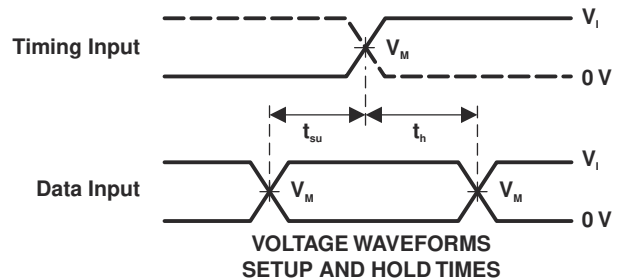
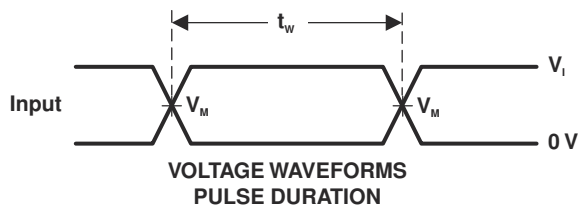
Figure 6-1. Load Circuit and Voltage Waveforms

(Open Drain)



TEST	S1
t_{PZL} (see Notes E and F)	V_{LOAD}
t_{PLZ} (see Notes E and G)	V_{LOAD}
t_{PHZ}/t_{PZH}	V_{LOAD}

V_{CC}	INPUTS		V_M	V_{LOAD}	C_L	R_L	V_{Δ}
	V_I	t_r/t_f					
$1.8 V \pm 0.15 V$	V_{CC}	$\leq 2 \text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	1 k Ω	0.15 V
$2.5 V \pm 0.2 V$	V_{CC}	$\leq 2 \text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	500 Ω	0.15 V
$3.3 V \pm 0.3 V$	3 V	$\leq 2.5 \text{ ns}$	1.5 V	6 V	50 pF	500 Ω	0.3 V
$5 V \pm 0.5 V$	V_{CC}	$\leq 2.5 \text{ ns}$	$V_{CC}/2$	$2 \times V_{CC}$	50 pF	500 Ω	0.3 V



- NOTES:
- C_L includes probe and jig capacitance.
 - Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
 - All input pulses are supplied by generators having the following characteristics: PRR $\leq 10 \text{ MHz}$, $Z_O = 50 \Omega$.
 - The outputs are measured one at a time, with one transition per measurement.
 - Because this device has open-drain outputs, t_{PLZ} and t_{PZL} are the same as t_{PD} .
 - t_{PZL} is measured at V_M .
 - t_{PLZ} is measured at $V_{OL} + V_{\Delta}$.
 - All parameters and waveforms are not applicable to all devices.

Figure 6-2. Load Circuit and Voltage Waveforms

7 Detailed Description

7.1 Overview

The SN74LVC1G38 device is a single two-input NAND gate with open-drain outputs designed for operation from 1.65V to 5.5V V_{CC} . The device performs the Boolean function $Y = \overline{A \times B}$ or $Y = \overline{A} + \overline{B}$ in positive logic.

This device is fully specified for partial-power-down applications using I_{off} . The I_{off} circuitry disables the outputs when the device is powered down. This inhibits current backflow into the device which prevents damage to the device.

7.2 Functional Block Diagram

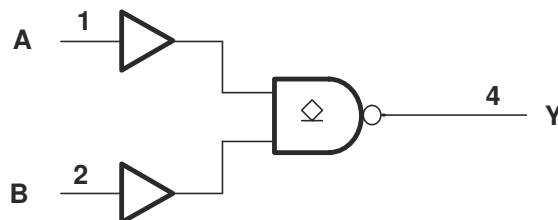


Figure 7-1. Logic Diagram (Positive Logic)

7.3 Feature Description

7.3.1 High-Drive Open-Drain Output

The open-drain output allows the device to sink current when the output is LOW and maintains a high impedance state when the output is HIGH. The high drive capability of this device creates fast edges into light loads so routing and load conditions must be considered to prevent ringing. Additionally, the outputs of this device are capable of driving larger currents than the device can sustain without being damaged. It is important for the power output of the device to be limited to avoid thermal runaway and damage due to over-current. The electrical and thermal limits defined in the [Absolute Maximum Ratings](#) must be followed at all times.

7.3.2 Standard CMOS Inputs

Standard CMOS inputs are high impedance and are typically modeled as a resistor in parallel with the input capacitance given in the [Electrical Characteristics](#). The worst case resistance is calculated with the maximum input voltage, given in the [Absolute Maximum Ratings](#), and the maximum input leakage current, given in the [Electrical Characteristics](#), using Ohm's law ($R = V \div I$).

Signals applied to the inputs need to have fast edge rates, as defined by $\Delta t/\Delta v$ in [Recommended Operating Conditions](#) to avoid excessive currents and oscillations. If a slow or noisy input signal is required, a device with a Schmitt-trigger input must be used to condition the input signal prior to the standard CMOS input.

7.3.3 Clamp Diodes

The inputs and outputs to this device have negative clamping diodes.

CAUTION

Voltages beyond the values specified in the [Absolute Maximum Ratings](#) table can cause damage to the device. The input negative-voltage and output voltage ratings may be exceeded if the input and output clamp-current ratings are observed.

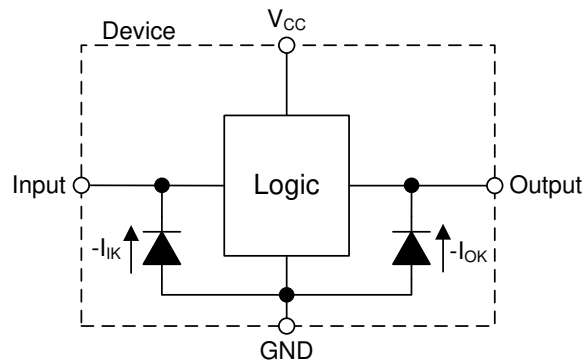


Figure 7-2. Electrical Placement of Clamping Diodes for Each Input and Output

7.3.4 Partial Power Down (I_{off})

The inputs and outputs for this device enter a high impedance state when the supply voltage is 0V. The maximum leakage into or out of any input or output pin on the device is specified by I_{off} in the .

7.3.5 Over-Voltage Tolerant Inputs

Input signals to this device can be driven above the supply voltage so long as they remain below the maximum input voltage value specified in the [Absolute Maximum Ratings](#).

7.3.6 Up Translation and Down Translation Capable Outputs

Outputs of this device can be driven above the supply voltage so long as they remain below the maximum output voltage value specified in the [Absolute Maximum Ratings](#). When the device is not actively driving LOW, the output is in the high impedance state. If a pull-up resistor is connected from the output to a power supply (of any valid value), the output will be driven by this supply, and therefore can have a voltage that is either higher or lower than the V_{CC} supply of the device. An application of this device performing up-translation is depicted in [Application and Implementation](#), where additional design details are provided.

7.4 Device Functional Modes

Table 7-1 lists the functional modes of the SN74LVC1G38 device.

Table 7-1. Function Table

INPUTS		OUTPUT Y
A	B	
L	L	Hi-Z
L	H	Hi-Z
H	L	Hi-Z
H	H	L

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

Open-drain devices are intrinsically capable of voltage translation. In this application, a 1.8V logic signal is inverted and up-translated to 5V at the output when the EN signal input is driven high by a 3.3V signal. The output is held at 5V in this scenario when the output of the device is in the high impedance state.

8.2 Typical Application

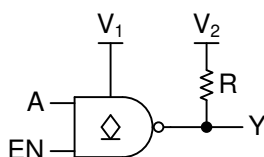


Figure 8-1. Gated Voltage Translating Inverter Schematic Using SN74LVC1G38

8.2.1 Design Requirements

The supply voltage at V_1 must be set to provide input thresholds for the signals A and EN. This device uses CMOS technology and has an open-drain output. Outputs of open-drain devices can be tied directly together to produce a wired-OR configuration. This device has high current drive that creates fast edges into light loads so routing and load conditions must be considered to prevent ringing.

8.2.2 Detailed Design Procedure

- Recommended Input Conditions
 - Rise time and fall time specs. See $(\Delta t/\Delta V)$ in the [Recommended Operating Conditions](#) table.
 - Specified high and low levels. See $(V_{IH}$ and $V_{IL})$ in the [Recommended Operating Conditions](#) table.
 - Inputs are overvoltage tolerant allowing them to go as high as $(V_I \text{ max})$ in the [Recommended Operating Conditions](#) table at any valid VCC.
- Recommended Output Conditions
 - Load currents should not exceed $(I_O \text{ max})$. These limits are located in the [Absolute Maximum Ratings](#) table.
 - Outputs can be pulled above VCC for up-translation applications as long as the maximum output voltage in the [Absolute Maximum Ratings](#) table is observed.

8.2.3 Application Curve

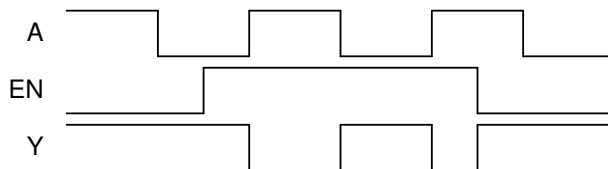


Figure 8-2. Application Timing Diagram

8.3 Power Supply Recommendations

The power supply can be any voltage between the MIN and MAX supply voltage rating located in the [Recommended Operating Conditions](#) table.

The V_{CC} pin must have a good bypass capacitor to prevent power disturbance. Parallel multiple bypass caps to reject different frequencies of noise is acceptable. $0.1\mu\text{F}$ and $1\mu\text{F}$ capacitors are commonly used in parallel. The bypass capacitor must be installed as close to the power pin as possible for best results.

8.4 Layout

8.4.1 Layout Guidelines

When using multiple bit logic devices, inputs must not float. In many cases, functions or parts of functions of digital logic devices are unused. Some examples are when only two inputs of a triple-input AND gate are used, or when only 3 of the 4-buffer gates are used. Such input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states.

Specified in [Figure 8-3](#) are rules that must be observed under all circumstances. All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally they are tied to GND or V_{CC} , whichever makes more sense or is more convenient.

Even low data rate digital signals can have high frequency signal components due to fast edge rates. When a printed-circuit board (PCB) trace turns a corner at a 90° angle, a reflection can occur. A reflection occurs primarily because of the change of width of the trace. At the apex of the turn, the trace width increases to 1.414 times the width. This increase upsets the transmission-line characteristics, especially the distributed capacitance and self-inductance of the trace which results in the reflection. Not all PCB traces can be straight and therefore some traces must turn corners. [Figure 8-4](#) shows progressively better techniques of rounding corners. Only the last example (BEST) maintains constant trace width and minimizes reflections.

8.4.2 Layout Example

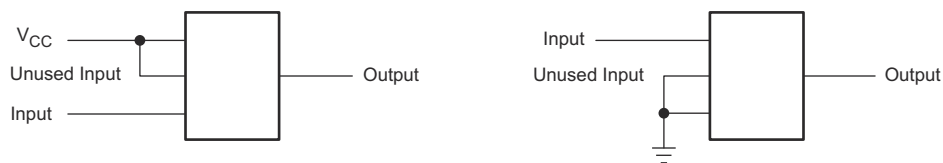


Figure 8-3. Proper Multi-Gate Input Termination Diagram

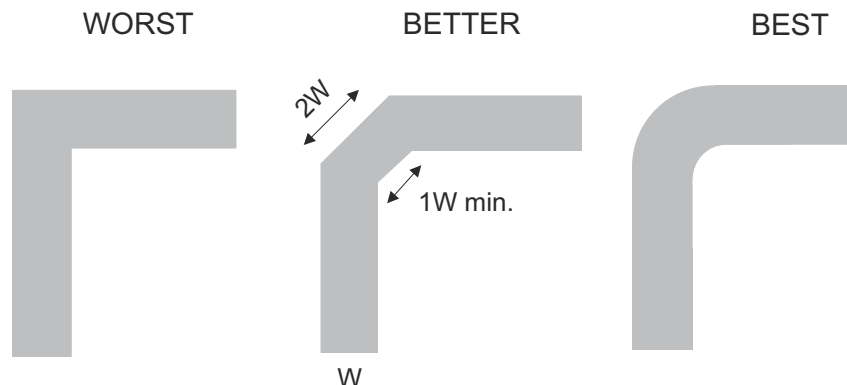


Figure 8-4. Trace Example

9 Device and Documentation Support

9.1 Documentation Support

9.1.1 Related Documentation

For related documentation see the following:

- Texas Instruments, [Designing and Manufacturing with TI's X2SON Packages Application Note](#)
- Texas Instruments, [How to Select Little Logic Application Note](#)
- Texas Instruments, [Implications of Slow or Floating CMOS Inputs Application Note](#)
- Texas Instruments, [Understanding and Interpreting Standard-Logic Data Sheets Application Note](#)
- Texas Instruments, [Introduction to Logic Application Note](#)
- Texas Instruments, [Signal Switch Data Book User's Guide](#)
- Texas Instruments, [LVC and LV Low-Voltage CMOS Logic Data Book User's Guide](#)
- Texas Instruments, [Low-Voltage Logic \(LVC\) Designer's Guide User's Guide](#)

9.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

9.3 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

9.4 Trademarks

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9.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.6 Glossary

[TI Glossary](#)

This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from February 1, 2020 to August 12, 2026 (from Revision G (February 2020) to Revision H (August 2026))

Page

• Updated <i>Device Information</i> table to include body and package dimensions.....	1
• Updated the numbering format for tables, figures, and cross-references throughout the document.....	1
• Added <i>Thermal Information</i> table.....	5
• Changed Junction-to-ambient thermal resistance value for DBV package from: 247.2°C/W to: 357.1°C/W.....	5
• Changed Junction-to-case (top) thermal resistance value for DBV package from: 154.5°C/W to: 263.7°C/W..	5
• Changed Junction-to-board characterization value for DBV package from: 86.8°C/W to: 264.4°C/W.....	5
• Changed Junction-to-top characterization value for DBV package from: 58°C/W to: 195.6°C/W.....	5
• Changed Junction-to-board characterization value for DBV package from: 86.4°C/W to: 262.2°C/W.....	5
• Changed Junction-to-ambient thermal resistance value for DCK package from: 276.1°C/W to: 371°C/W.....	5
• Changed Junction-to-case (top) thermal resistance value for DCK package from: 178.9°C/W to: 297.5°C/W..	5
• Changed Junction-to-board characterization value for DCK package from: 70.9°C/W to: 258.6°C/W.....	5
• Changed Junction-to-top characterization value for DCK package from: 47°C/W to: 195.6°C/W.....	5
• Changed Junction-to-board characterization value for DCK package from: 69.3°C/W to: 256.2°C/W.....	5

Changes from Revision F (October 2017) to Revision G (February 2020)

Page

• Changed the T _A high-end temp from +1255°C to +125°C in the Test Conditions field of the Electrical Characteristics table.	6
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11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74LVC1G38DBVR	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 125	(C385, C38F, C38J, C38R) (C38H, C38P, C38S)
SN74LVC1G38DBVR.A	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(C385, C38F, C38J, C38R) (C38H, C38P, C38S)
SN74LVC1G38DBVR.B	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(C385, C38F, C38J, C38R) (C38H, C38P, C38S)
SN74LVC1G38DBVRE4	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C385 C38S
SN74LVC1G38DBVRG4	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C385 C38S
SN74LVC1G38DBVRG4.B	Active	Production	SOT-23 (DBV) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C385 C38S
SN74LVC1G38DBVT	Active	Production	SOT-23 (DBV) 5	250 SMALL T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 125	(C385, C38J, C38R) (C38H, C38S)
SN74LVC1G38DBVT.B	Active	Production	SOT-23 (DBV) 5	250 SMALL T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(C385, C38J, C38R) (C38H, C38S)
SN74LVC1G38DBVTG4	Active	Production	SOT-23 (DBV) 5	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C385 C38S
SN74LVC1G38DBVTG4.B	Active	Production	SOT-23 (DBV) 5	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	C385 C38S
SN74LVC1G38DCKR	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 125	(D75, D7F, D7J, D7R) (D7H, D7P, D7S)
SN74LVC1G38DCKR.A	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(D75, D7F, D7J, D7R) (D7H, D7P, D7S)

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74LVC1G38DCKR.B	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(D75, D7F, D7J, D7R) (D7H, D7P, D7S)
SN74LVC1G38DCKRG4	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	D75 D7S
SN74LVC1G38DCKRG4.A	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	D75 D7S
SN74LVC1G38DCKRG4.B	Active	Production	SC70 (DCK) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	D75 D7S
SN74LVC1G38DCKT	Active	Production	SC70 (DCK) 5	250 SMALL T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 125	(D75, D7J, D7R) (D7H, D7S)
SN74LVC1G38DCKT.B	Active	Production	SC70 (DCK) 5	250 SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(D75, D7J, D7R) (D7H, D7S)
SN74LVC1G38DPWR	Active	Production	X2SON (DPW) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	BP
SN74LVC1G38DPWR.B	Active	Production	X2SON (DPW) 5	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	BP
SN74LVC1G38DRYR	Active	Production	SON (DRY) 6	5000 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	D7
SN74LVC1G38DRYR.B	Active	Production	SON (DRY) 6	5000 LARGE T&R	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	D7
SN74LVC1G38DSFR	Active	Production	SON (DSF) 6	5000 LARGE T&R	Yes	NIPDAU NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	D7
SN74LVC1G38DSFR.B	Active	Production	SON (DSF) 6	5000 LARGE T&R	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	D7
SN74LVC1G38DSFRG4	Active	Production	SON (DSF) 6	5000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	D7
SN74LVC1G38DSFRG4.B	Active	Production	SON (DSF) 6	5000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	D7

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

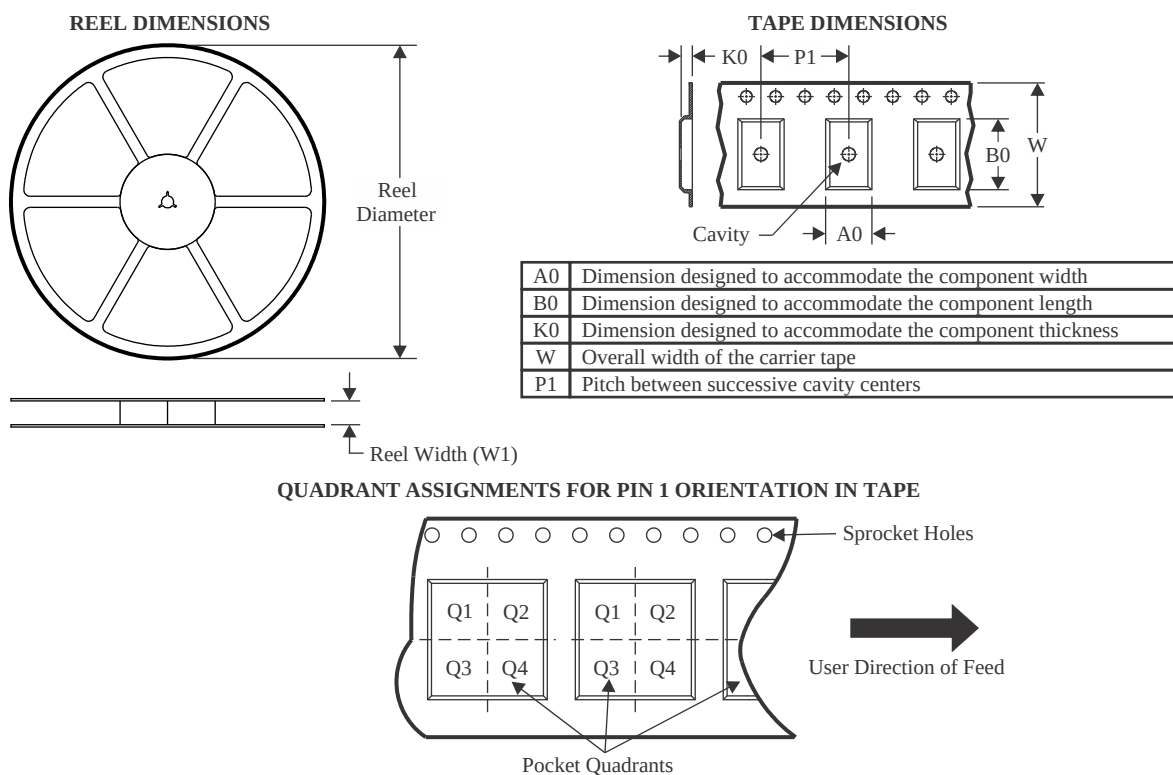
⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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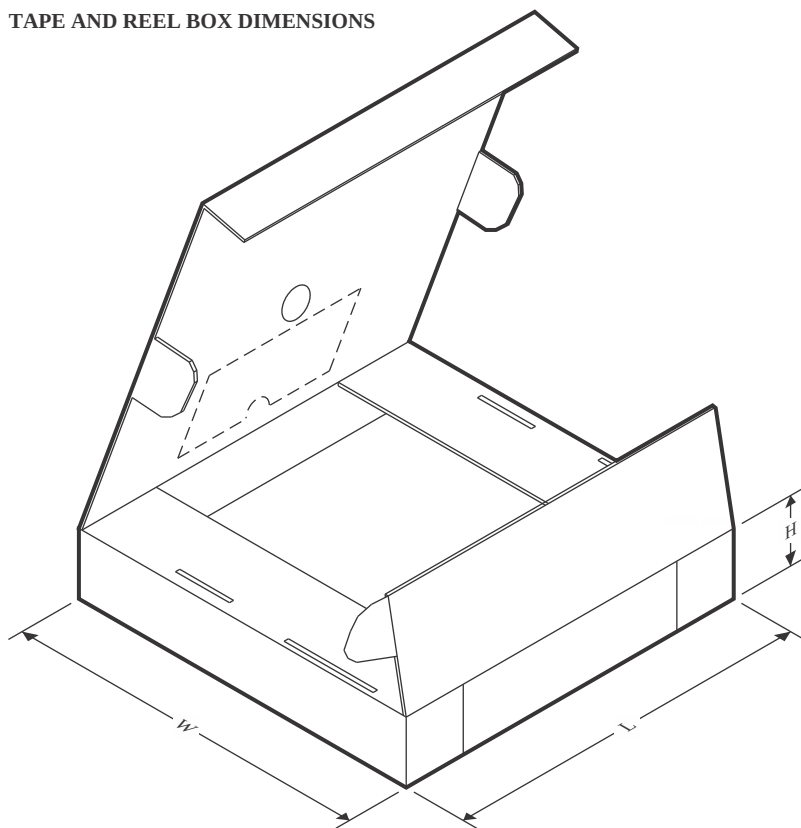
TAPE AND REEL INFORMATION


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	178.0	9.2	3.3	3.23	1.55	4.0	8.0	Q3
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	178.0	9.0	3.3	3.2	1.4	4.0	8.0	Q3
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	178.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G38DBVRG4	SOT-23	DBV	5	3000	178.0	9.2	3.3	3.23	1.55	4.0	8.0	Q3
SN74LVC1G38DBVT	SOT-23	DBV	5	250	178.0	9.2	3.3	3.23	1.55	4.0	8.0	Q3
SN74LVC1G38DBVT	SOT-23	DBV	5	250	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74LVC1G38DBVT	SOT-23	DBV	5	250	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
SN74LVC1G38DBVT	SOT-23	DBV	5	250	178.0	9.0	3.3	3.2	1.4	4.0	8.0	Q3
SN74LVC1G38DBVTG4	SOT-23	DBV	5	250	178.0	9.2	3.3	3.23	1.55	4.0	8.0	Q3
SN74LVC1G38DCKR	SC70	DCK	5	3000	180.0	8.4	2.3	2.55	1.2	4.0	8.0	Q3
SN74LVC1G38DCKR	SC70	DCK	5	3000	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G38DCKR	SC70	DCK	5	3000	180.0	8.4	2.47	2.3	1.25	4.0	8.0	Q3
SN74LVC1G38DCKR	SC70	DCK	5	3000	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G38DCKR	SC70	DCK	5	3000	178.0	8.4	2.25	2.45	1.2	4.0	8.0	Q3

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74LVC1G38DCKRG4	SC70	DCK	5	3000	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G38DCKT	SC70	DCK	5	250	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
SN74LVC1G38DCKT	SC70	DCK	5	250	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74LVC1G38DCKT	SC70	DCK	5	250	180.0	8.4	2.47	2.3	1.25	4.0	8.0	Q3
SN74LVC1G38DPWR	X2SON	DPW	5	3000	178.0	8.4	0.91	0.91	0.5	2.0	8.0	Q3
SN74LVC1G38DRYR	SON	DRY	6	5000	180.0	9.5	1.15	1.6	0.75	4.0	8.0	Q1
SN74LVC1G38DRYR	SON	DRY	6	5000	180.0	8.4	1.25	1.6	0.7	4.0	8.0	Q1
SN74LVC1G38DSFR	SON	DSF	6	5000	180.0	9.5	1.16	1.16	0.5	4.0	8.0	Q2
SN74LVC1G38DSFR	SON	DSF	6	5000	180.0	8.4	1.16	1.16	0.63	4.0	8.0	Q2
SN74LVC1G38DSFRG4	SON	DSF	6	5000	180.0	9.5	1.16	1.16	0.5	4.0	8.0	Q2

TAPE AND REEL BOX DIMENSIONS



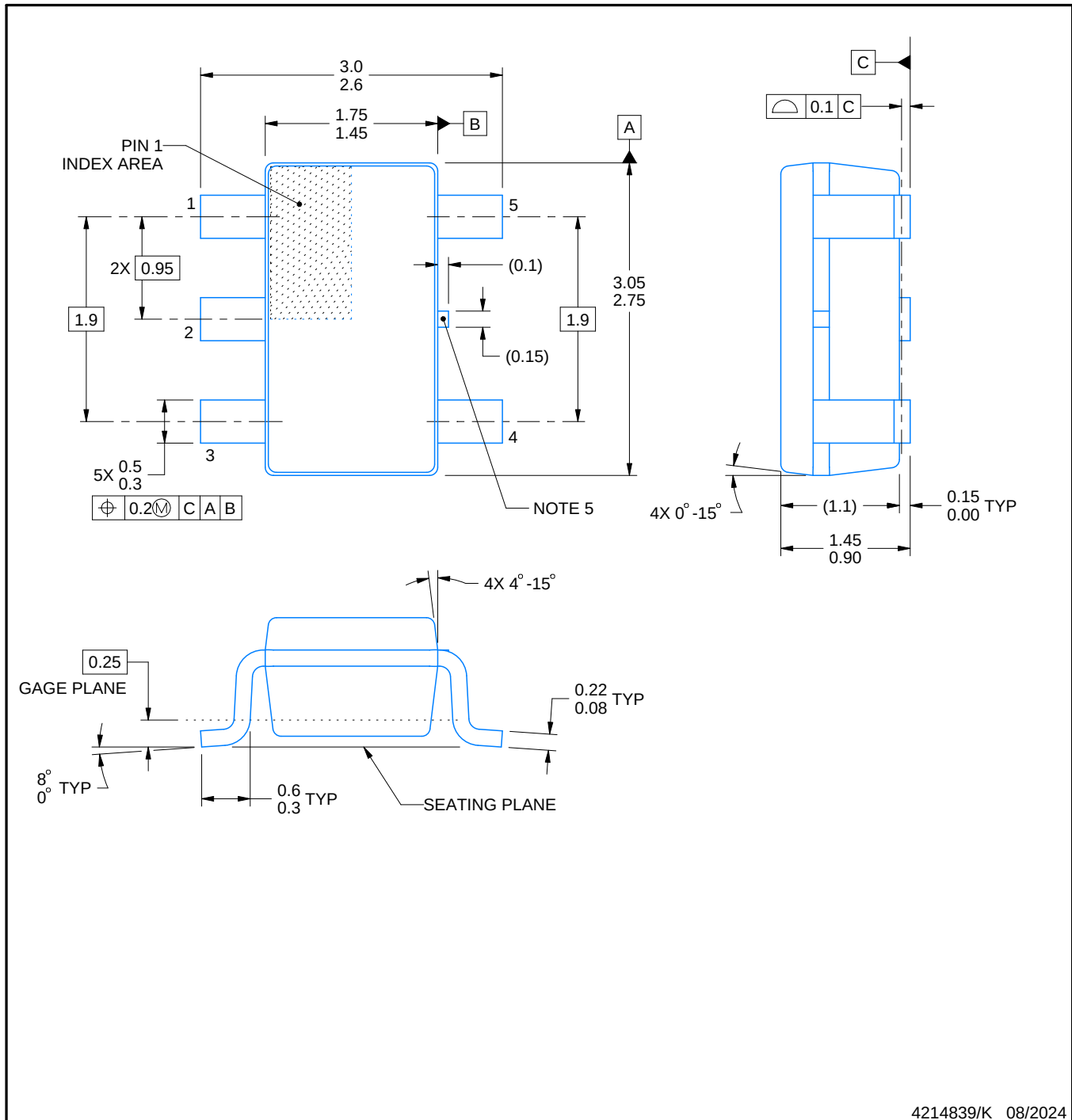
*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	202.0	201.0	28.0
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	210.0	185.0	35.0
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G38DBVR	SOT-23	DBV	5	3000	208.0	191.0	35.0
SN74LVC1G38DBVRG4	SOT-23	DBV	5	3000	180.0	180.0	18.0
SN74LVC1G38DBVT	SOT-23	DBV	5	250	180.0	180.0	18.0
SN74LVC1G38DBVT	SOT-23	DBV	5	250	210.0	185.0	35.0
SN74LVC1G38DBVT	SOT-23	DBV	5	250	202.0	201.0	28.0
SN74LVC1G38DBVT	SOT-23	DBV	5	250	180.0	180.0	18.0
SN74LVC1G38DBVTG4	SOT-23	DBV	5	250	180.0	180.0	18.0
SN74LVC1G38DCKR	SC70	DCK	5	3000	210.0	185.0	35.0
SN74LVC1G38DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G38DCKR	SC70	DCK	5	3000	202.0	201.0	28.0
SN74LVC1G38DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G38DCKR	SC70	DCK	5	3000	208.0	191.0	35.0
SN74LVC1G38DCKRG4	SC70	DCK	5	3000	180.0	180.0	18.0
SN74LVC1G38DCKT	SC70	DCK	5	250	180.0	180.0	18.0

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74LVC1G38DCKT	SC70	DCK	5	250	180.0	180.0	18.0
SN74LVC1G38DCKT	SC70	DCK	5	250	202.0	201.0	28.0
SN74LVC1G38DPWR	X2SON	DPW	5	3000	205.0	200.0	33.0
SN74LVC1G38DRYR	SON	DRY	6	5000	184.0	184.0	19.0
SN74LVC1G38DRYR	SON	DRY	6	5000	202.0	201.0	28.0
SN74LVC1G38DSFR	SON	DSF	6	5000	184.0	184.0	19.0
SN74LVC1G38DSFR	SON	DSF	6	5000	202.0	201.0	28.0
SN74LVC1G38DSFRG4	SON	DSF	6	5000	184.0	184.0	19.0

DBV0005A**PACKAGE OUTLINE****SOT-23 - 1.45 mm max height**

SMALL OUTLINE TRANSISTOR



4214839/K 08/2024

NOTES:

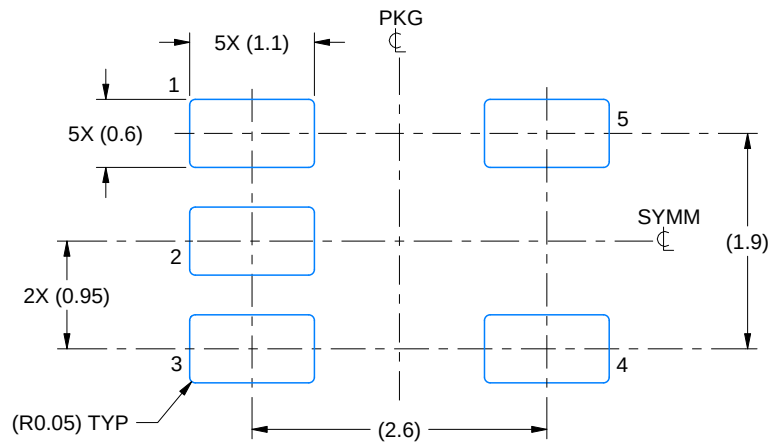
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC MO-178.
4. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25 mm per side.
5. Support pin may differ or may not be present.

EXAMPLE BOARD LAYOUT

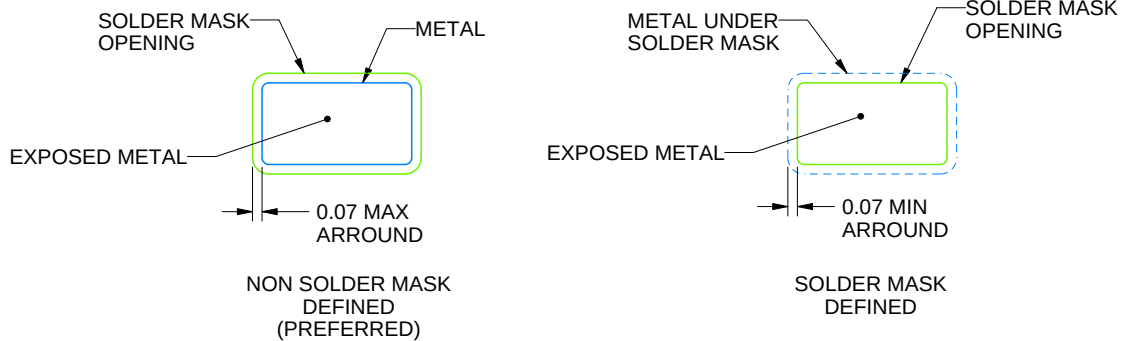
DBV0005A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X



SOLDER MASK DETAILS

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NOTES: (continued)

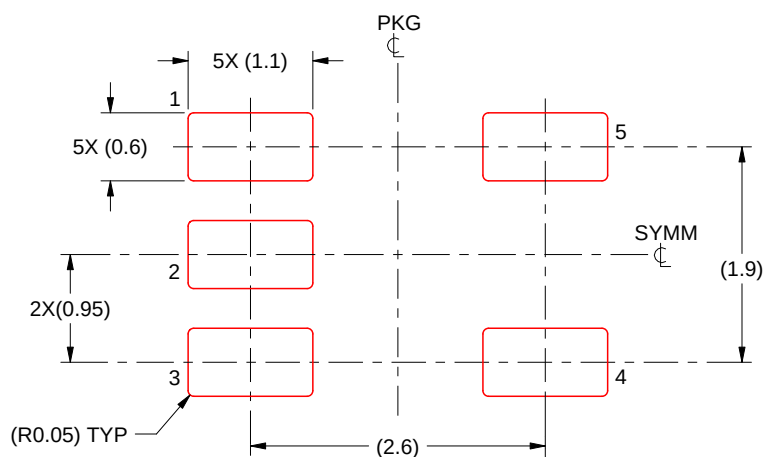
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DBV0005A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214839/K 08/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

GENERIC PACKAGE VIEW

DRY 6

USON - 0.6 mm max height

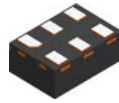
PLASTIC SMALL OUTLINE - NO LEAD



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4207181/G

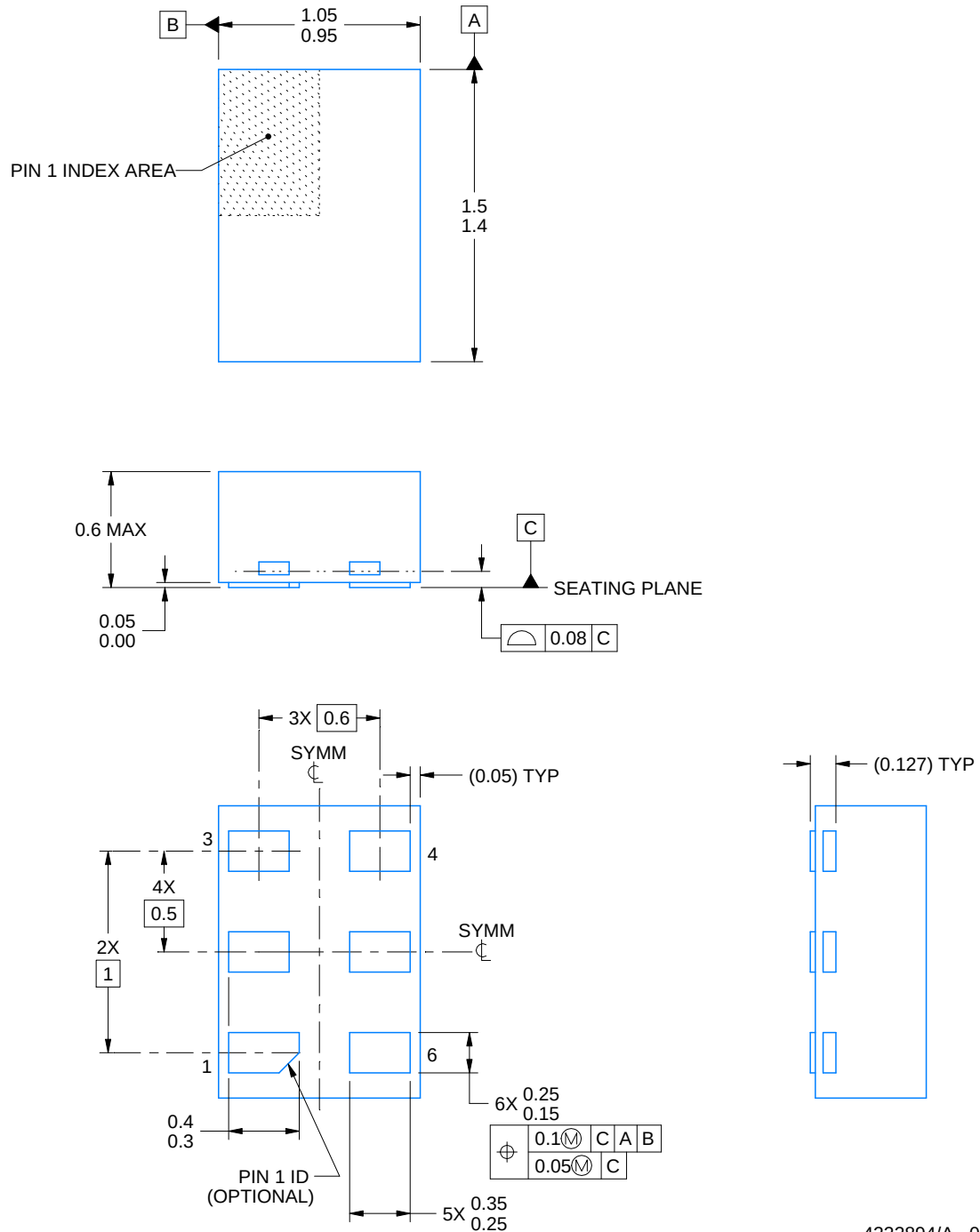
DRY0006A



PACKAGE OUTLINE

USON - 0.6 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



4222894/A 01/2018

NOTES:

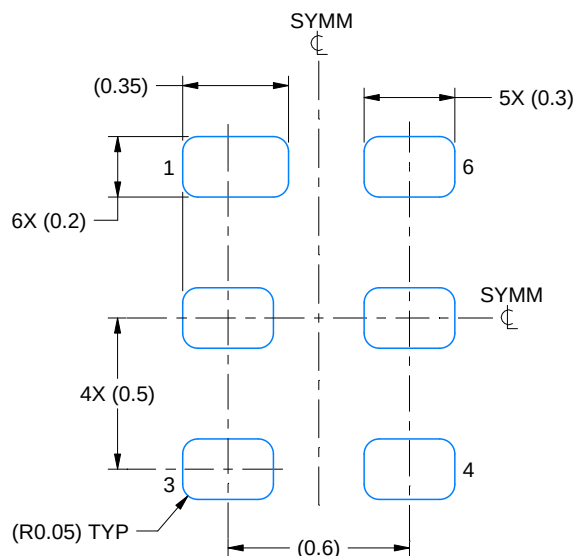
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.

EXAMPLE BOARD LAYOUT

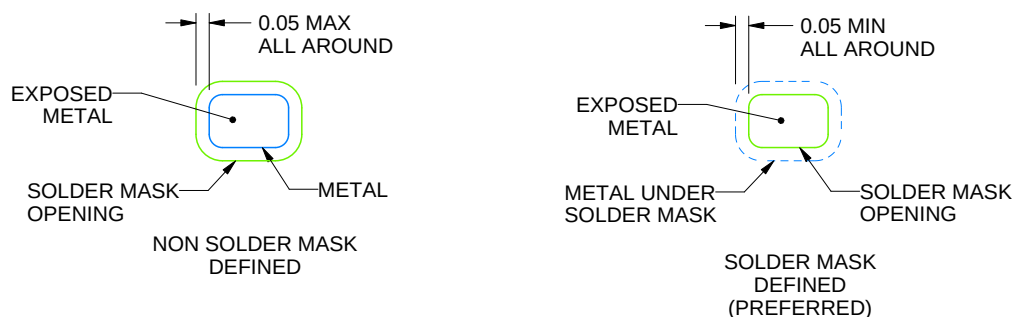
DRY0006A

USON - 0.6 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



LAND PATTERN EXAMPLE
1:1 RATIO WITH PKG SOLDER PADS
EXPOSED METAL SHOWN
SCALE:40X



SOLDER MASK DETAILS

4222894/A 01/2018

NOTES: (continued)

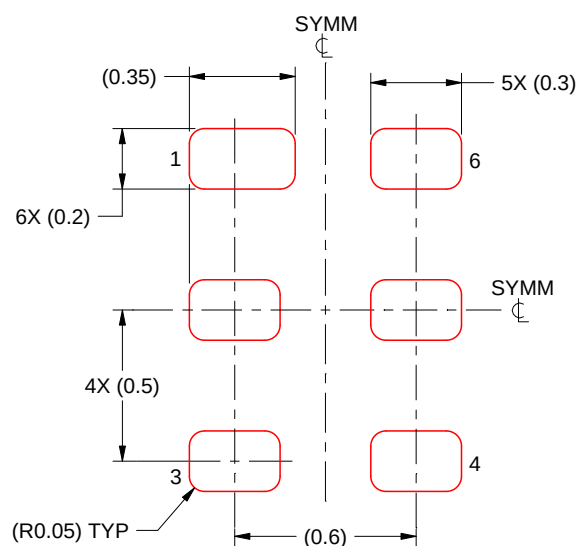
3. For more information, see QFN/SON PCB application report in literature No. SLUA271 (www.ti.com/lit/sluea271).

EXAMPLE STENCIL DESIGN

DRY0006A

USON - 0.6 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



SOLDER PASTE EXAMPLE
BASED ON 0.075 - 0.1 mm THICK STENCIL
SCALE:40X

4222894/A 01/2018

NOTES: (continued)

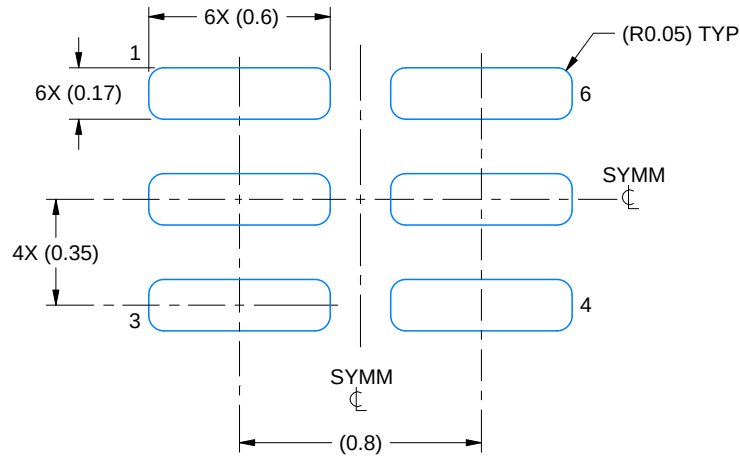
4. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

EXAMPLE BOARD LAYOUT

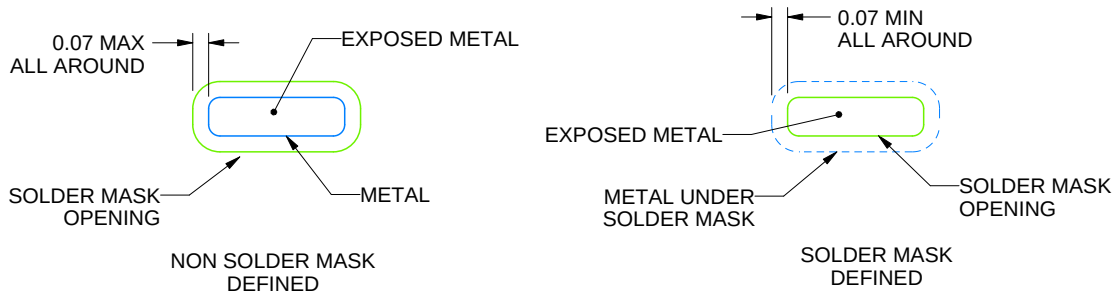
DSF0006A

X2SON - 0.4 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:40X



SOLDER MASK DETAILS

4220597/B 06/2022

NOTES: (continued)

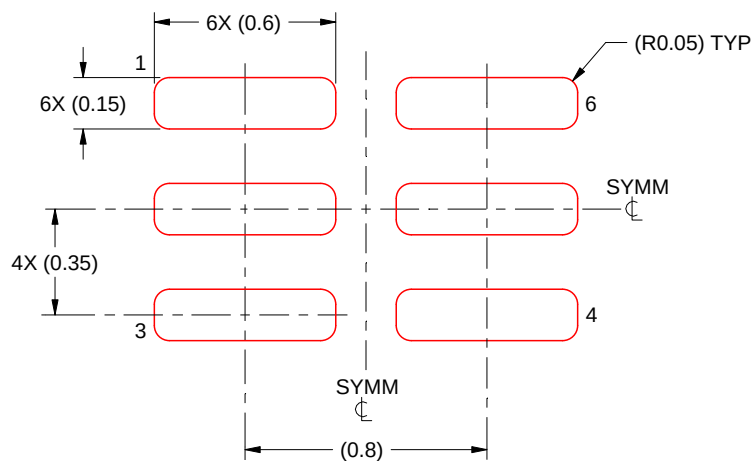
4. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/sluea271).

EXAMPLE STENCIL DESIGN

DSF0006A

X2SON - 0.4 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



SOLDER PASTE EXAMPLE
BASED ON 0.09 mm THICK STENCIL

PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE
SCALE:40X

4220597/B 06/2022

4. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

GENERIC PACKAGE VIEW

DPW 5

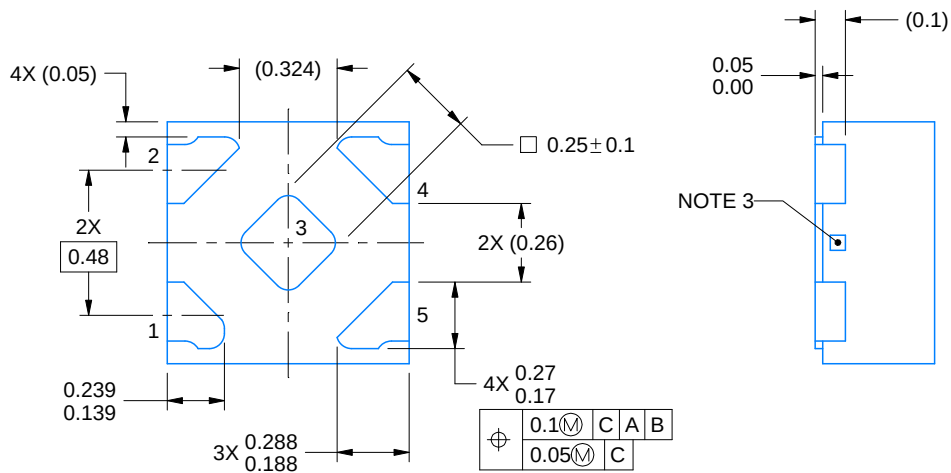
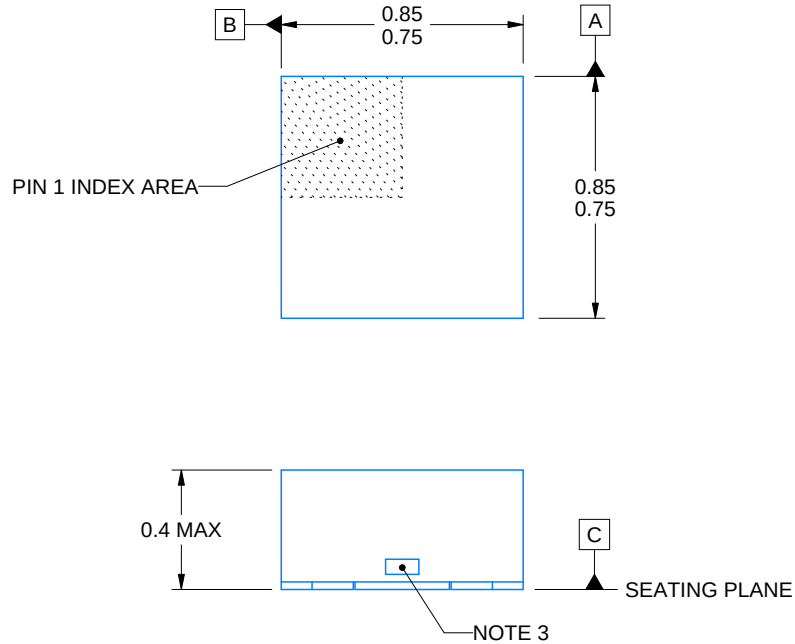
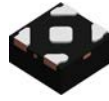
X2SON - 0.4 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4211218-3/D



4223102/D 03/2022

NOTES:

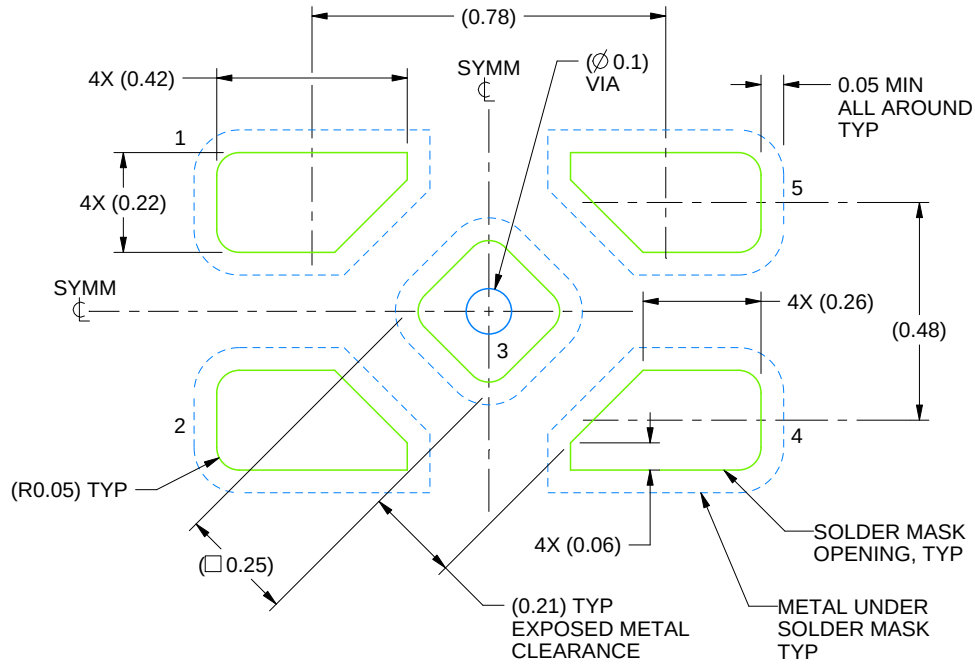
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The size and shape of this feature may vary.

EXAMPLE BOARD LAYOUT

DPW0005A

X2SON - 0.4 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



LAND PATTERN EXAMPLE
SOLDER MASK DEFINED
SCALE:60X

4223102/D 03/2022

NOTES: (continued)

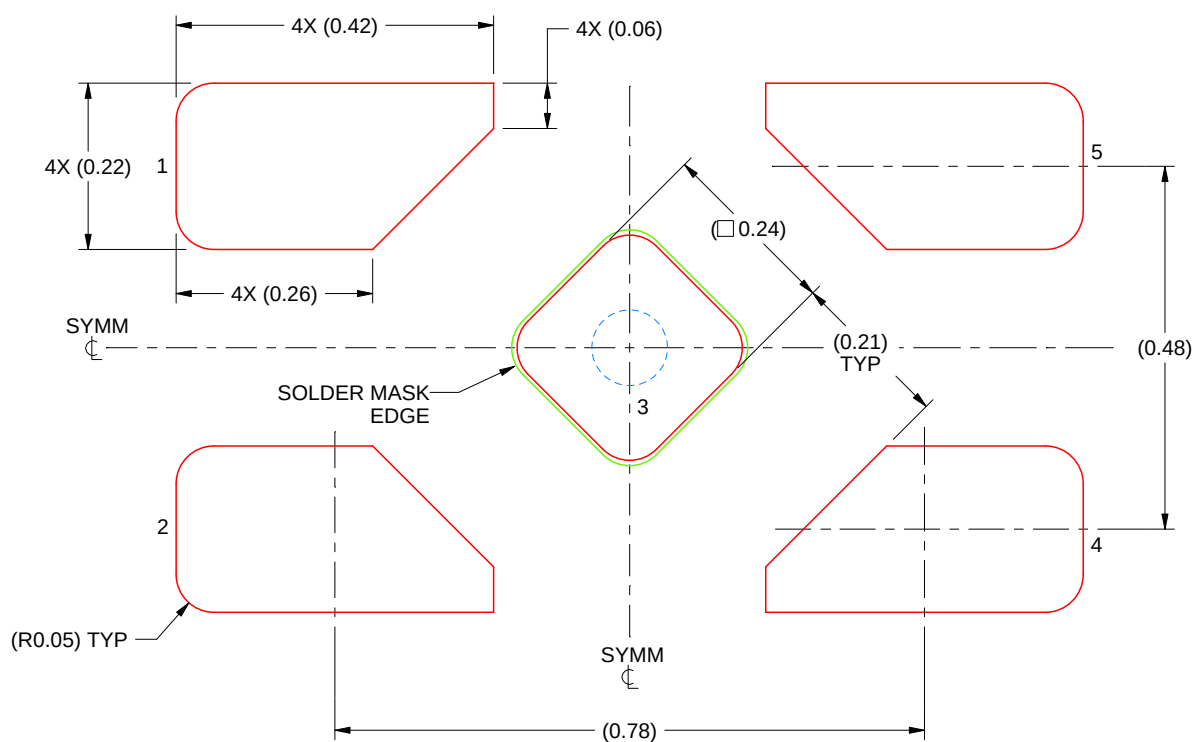
- This package is designed to be soldered to a thermal pad on the board. For more information, refer to QFN/SON PCB application note in literature No. SLUA271 (www.ti.com/lit/sluea271).

EXAMPLE STENCIL DESIGN

DPW0005A

X2SON - 0.4 mm max height

PLASTIC SMALL OUTLINE - NO LEAD



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL

EXPOSED PAD 3
92% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE
SCALE:100X

4223102/D 03/2022

NOTES: (continued)

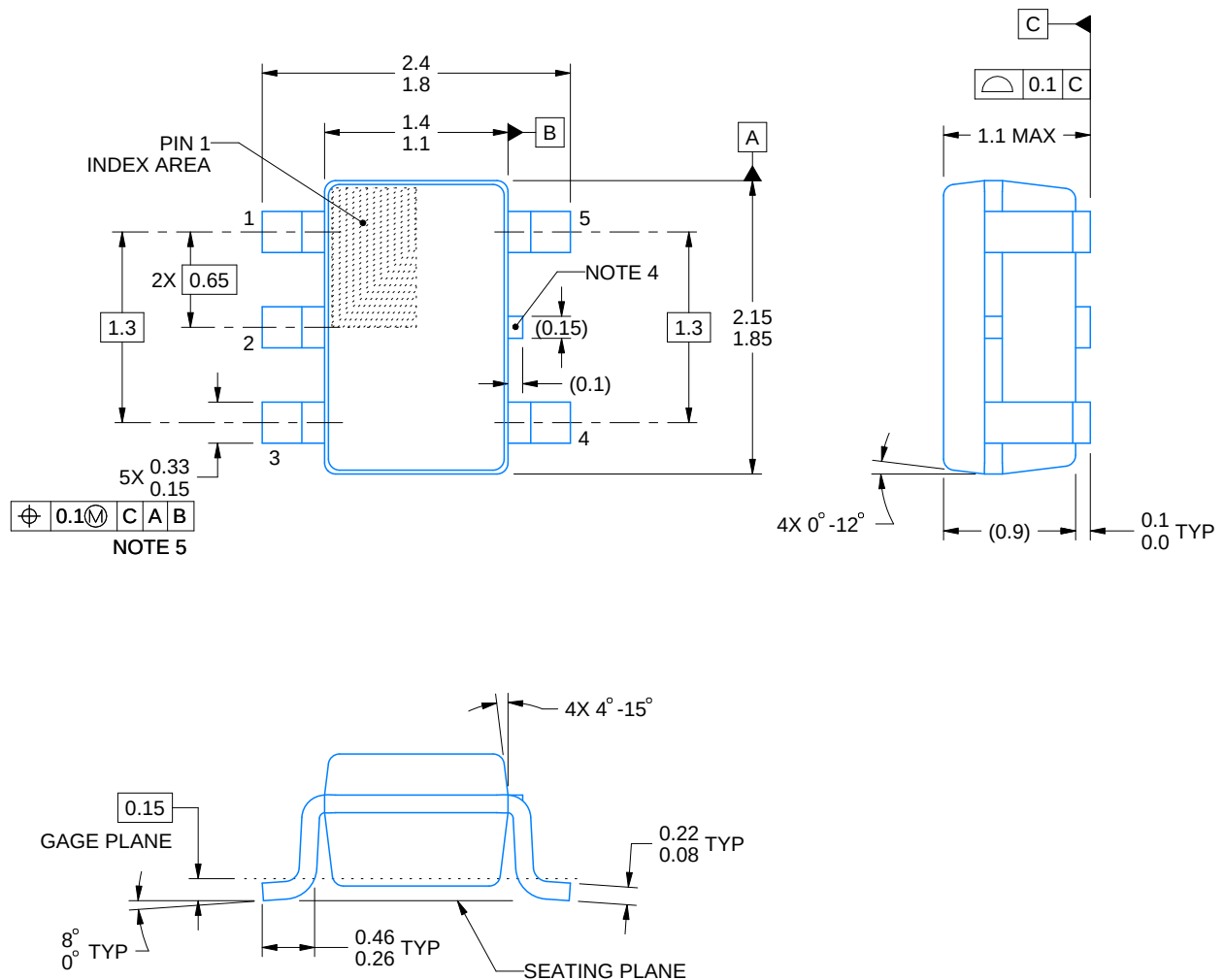
5. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

DCK0005A

PACKAGE OUTLINE

SOT - 1.1 max height

SMALL OUTLINE TRANSISTOR



4214834/G 11/2024

NOTES:

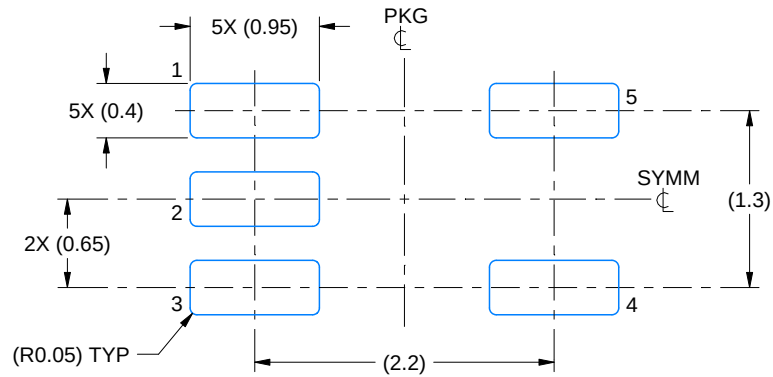
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC MO-203.
4. Support pin may differ or may not be present.
5. Lead width does not comply with JEDEC.
6. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25mm per side

EXAMPLE BOARD LAYOUT

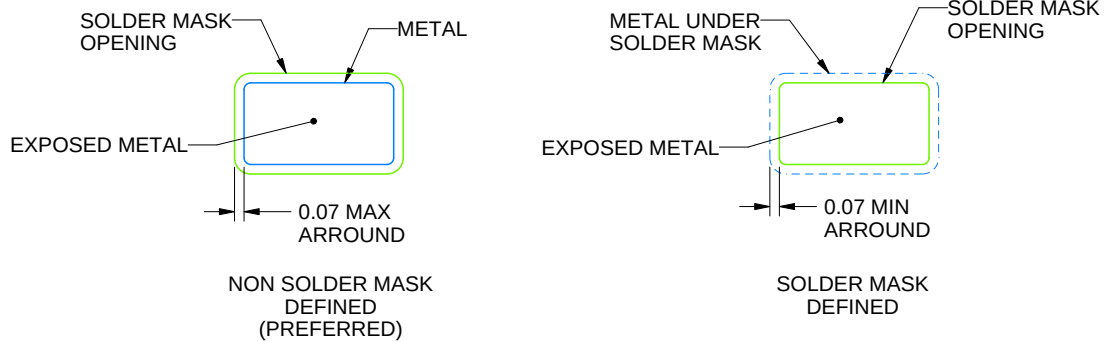
DCK0005A

SOT - 1.1 max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

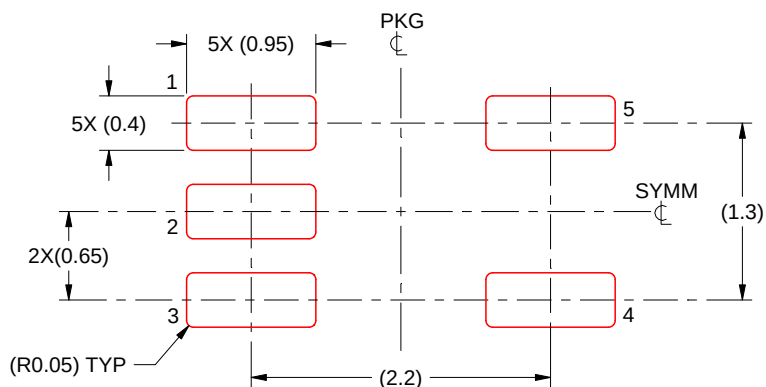


SOLDER MASK DETAILS

4214834/G 11/2024

NOTES: (continued)

7. Publication IPC-7351 may have alternate designs.
8. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 THICK STENCIL
SCALE:18X

4214834/G 11/2024

NOTES: (continued)

9. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
10. Board assembly site may have different recommendations for stencil design.

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